

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE: CSE4125 (Distributed Database Systems)
 Spring 2018 | Quiz – 2 | Marks 20 | Time: 40 Minutes

ID: _____ NAME: _____ SECTION: _____

Q-1) Consider the following global, fragmentation, and allocation schemata:

Global schema: **STUDENT** (*NUMBER, NAME, DEPT, SEM*)

Fragmentation schema: **STUDENT**₁ = **SL** *SEM* ≤ 4 and *DEPT* = "CSE" **STUDENT**

STUDENT₂ = **SL** *SEM* > 4 and *DEPT* = "CSE" **STUDENT**

STUDENT₃ = **SL** *SEM* > 4 and *DEPT* = "EEE" **STUDENT**

STUDENT₄ = **SL** *SEM* ≤ 4 and *DEPT* = "EEE" **STUDENT**

Allocation schema: **STUDENT**₁ at sites 1, 2 **STUDENT**₂ at sites 3, 4

STUDENT₃ at sites 5, 6 **STUDENT**₄ at sites 7, 8

(Assume that "EEE" and "CSE" are the only possible values for *DEPT*)

[12 marks] Write an application at level 3 of transparency that requires a student number from the terminal, and moves that student number from department "EEE" to "CSE" if he/she is enrolled in 1st semester.

Q-2) Consider the following fragments **ACCOUNT**₁ and **ACCOUNT**₂.

ACCOUNT₁			
<i>ACCNO</i>	<i>ACCTYPE</i>	<i>BALANCE</i>	<i>BRANCH</i>
3001	Savings	10,100	Dhaka
3002	Savings	15,020	Chittagong
3003	Savings	8,000	Chittagong
3004	Savings	12,000	Dhaka
3005	Savings	9,005	Khulna

ACCOUNT₂			
<i>ACCNO</i>	<i>ACCTYPE</i>	<i>BALANCE</i>	<i>BRANCH</i>
2001	Current	1,200	Dhaka
2002	Current	5,000	Khulna
2003	Current	4,080	Chittagong
2004	Current	2,000	Dhaka

Set of simple predicates **Pr** to obtain the fragments is given below.

P1: *ACCTYPE* = 'Savings'

P2: *ACCTYPE* = 'Current'

P3: *BALANCE* > 7,000

P4: *BALANCE* ≤ 7,000

Now, consider the following two applications –

(a) select * from **ACCOUNT**₁ where *BRANCH* = 'Dhaka' or *BRANCH* = 'Chittagong' or *BRANCH* = 'Khulna'

(b) select * from **ACCOUNT**₁ where *BALANCE* > 7,000

[08 marks] Determine if **Pr** is complete or not. Show reason behind your answer.